

## 1.4 continuity

To Define Continuity @ a point:

(all 3 must be true, has to be exact)

- 1)  $f(c)$  is defined NE: {VA, hole}
- 2)  $\lim_{x \rightarrow c} f(x)$  must exist NE: {jump, VA}
- 3)  $f(c) = \lim_{x \rightarrow c} f(x)$  NE: 

Practice 1.4:

@  $x=1$

$$f(x) = \begin{cases} 3x^2 + 4x - 5; & x > 1 \\ x + 1; & x < 1 \end{cases}$$

$f(1)$  is not defined, so  $f(x)$  is

not cont. at  $x=1$

@  $x=3$

$$g(x) = \begin{cases} 4x - 2; & x \leq 3 \\ 3 - x^2; & x > 3 \end{cases}$$

$g(3) = 10$ , but  $\lim_{x \rightarrow 3} g(x) \neq \text{DNE}$ , so

$g(x)$  is not continuous at  $x=3$

@  $x=3$

$$i(x) = \begin{cases} x^2 - 2x + 2; & x < 3 \\ 5; & x = 3 \\ 5x - 10; & x > 3 \end{cases}$$

$i(3) = 5$  ( $\lim_{x \rightarrow 3^+} i(x) = 5$ ) ( $\lim_{x \rightarrow 3^-} i(x) = 5$ ), and

$i(3) = \lim_{x \rightarrow 3} i(x) = 5$ , so  $i(x)$  is

continuous at  $x=3$

Doesn't have to be piecewise to be discontinuous\*

↓

make it continuous

$$f(x) = \begin{cases} cx^2 - 3 & x \leq 2 \\ cx + 2 & x > 2 \end{cases}$$

$$\lim_{x \rightarrow 2^-} (cx^2 - 3) = \lim_{x \rightarrow 2^+} (cx + 2)$$

$$4c - 3 = 2c + 2$$

$$c = 5/2$$

$$f(x) = \begin{cases} ax + b & x > 3 \\ 2bx - a & x < 3 \\ 2 & x = 3 \end{cases}$$

$$\lim_{x \rightarrow 3^-} 2bx - a = 2 \quad \lim_{x \rightarrow 3^+} ax + b = 2$$

$$6b - a$$

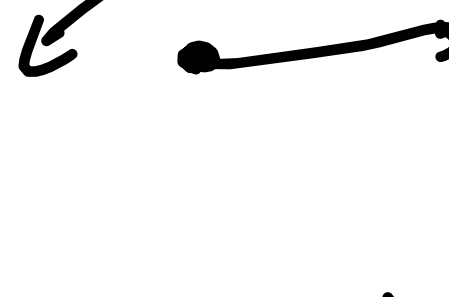
$$ax + 6$$

## Discontinuity

1. Holes: Point discontinuity

$\frac{x^2 - 4}{x - 2}$  Removable discontinuity

2. Step: Non-removable discontinuity



3. VA: asymptote discontinuity (non-removal)

